

Original Article

Effect of radiation dosage to vital ocular structures on visual outcomes: Re-evaluation for the future

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ABSTRACT

OBJECTIVE: To explore the impact of radiation dosage on vital ocular structures to identify target dosage for vision preservation in patients with uveal melanoma treated with Eye Physics plaque brachytherapy.

DESIGN: Retrospective study.

SUBJECTS: Patients with choroidal melanoma treated with I-125 brachytherapy between 2017 and 2021 with at least 1-year follow up.

METHODS: Retrospective chart review was performed. Time-to-event analysis was used to evaluate the effect of total dosage on ophthalmic structures. Other parameters analyzed include initial visual acuity, presence or absence of radiation retinopathy, tumor thickness, and tumor proximity to the fovea and optic nerve. Evaluation was based on 2 events: final best visual acuity of $\leq 20/200$ and visual loss ≥ 5 lines from baseline. Univariate and multivariate Cox proportional hazards models were used for analysis.

MAIN OUTCOME MEASURES: Visual acuity at baseline and each follow up visit.

RESULTS: One hundred thirty-six patients met inclusion criteria. Initial visual acuity was 20/20–20/40 in 71%, and $\leq 20/200$ in 7% of patients. At final visit, 27% patients had 20/20–20/40 and 50% had $\leq 20/200$. Higher total dose and average dose rate to the optic disc, fovea, lens, opposite retina and sclera were associated with worse visual outcomes. Foveal dosage < 35 Gy resulted in the greatest proportion of patients maintaining useful vision followed by a dose of < 27.5 Gy to the optic nerve.

CONCLUSIONS: < 35 Gy to the fovea and < 27.5 Gy to the optic disc results in $> 50\%$ of patients maintaining useful vision and may represent dose target values and are important when considering neoadjuvant treatments prior to definitive plaque brachytherapy. © 2025 American Brachytherapy Society. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords:

Brachytherapy; Uveal melanoma; Visual outcomes; Radiation dosage

Plaque brachytherapy is a highly successful, globe-salvaging treatment option for uveal melanoma (1). It was first introduced by Deutschman in 1915, who implanted

radium-228 capsules in fractionated doses, and was then popularized in the 1960's by Stallard, who used Cobalt-60 (2,3). The Collaborative Ocular Melanoma Study (COMS), performed between 1987 and 1998, was the first and only study to date evaluating the use of plaque brachytherapy in a randomized controlled trial and proved the noninferiority of Iodine-125 plaque brachytherapy compared with enucleation. This led to a paradigm shift in treatment, and plaque brachytherapy is now the most widely used treatment modality for uveal melanoma (4–8).

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